

Vectors and Vector-Valued Functions

ANSWER KEY



Vector operations in 3D

- 1 Let $\mathbf{a} = \langle 2, -1, 3 \rangle$ and $\mathbf{b} = \langle 1, 4, -2 \rangle$. Find:

a $\mathbf{a} + \mathbf{b} = \langle 3, 3, 1 \rangle$ **b** $\mathbf{a} \cdot \mathbf{b} = 2 - 4 - 6 = -8$ **c** $|\mathbf{a}| = \sqrt{14}$

- 2 Find the cross product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 1, 0, 2 \rangle$ and $\mathbf{b} = \langle 3, 1, -1 \rangle$.

$\mathbf{a} \times \mathbf{b} = \langle (0)(-1) - (2)(1), (2)(3) - (1)(-1), (1)(1) - (0)(3) \rangle = \langle -2, 7, 1 \rangle$.

Vector-valued functions: velocity and acceleration

- 3 A particle's position is $\mathbf{r}(t) = \langle t^2, 3t, t^3 \rangle$.

a Find the velocity vector $\mathbf{v}(t)$. $\mathbf{v}(t) = \langle 2t, 3, 3t^2 \rangle$.

b Find the acceleration vector $\mathbf{a}(t)$. $\mathbf{a}(t) = \langle 2, 0, 6t \rangle$.

c Find the speed $|\mathbf{v}(1)|$. $\mathbf{v}(1) = \langle 2, 3, 3 \rangle$, speed $= \sqrt{4 + 9 + 9} = \sqrt{22}$.

Unit tangent vectors and arc length

- 4 For $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$, find $\mathbf{r}'(t)$ and the speed $|\mathbf{r}'(t)|$.

$\mathbf{r}'(t) = \langle -3\sin t, 3\cos t, 4 \rangle$. $|\mathbf{r}'(t)| = \sqrt{9\sin^2 t + 9\cos^2 t + 16} = \sqrt{9 + 16} = 5$ (constant speed).

- 5 Find the arc length of $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \leq t \leq 2\pi$, using the constant speed found above.

$L = \int_0^{2\pi} |\mathbf{r}'(t)| dt = \int_0^{2\pi} 5 dt = 10\pi$.

Applications of the cross product

- 6 Find a vector orthogonal to both $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, using the cross product.

$\mathbf{u} \times \mathbf{v} = \langle (2)(3) - (0)(1), (0)(0) - (1)(3), (1)(1) - (2)(0) \rangle = \langle 6, -3, 1 \rangle$.

- 7 Find the area of the parallelogram spanned by $\mathbf{u} = \langle 2, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 3, 0 \rangle$, using $|\mathbf{u} \times \mathbf{v}|$.

$\mathbf{u} \times \mathbf{v} = \langle 0, 0, 6 \rangle$. Area $= |\langle 0, 0, 6 \rangle| = 6$.

Lines and planes in space

- 8 Write parametric equations for the line through $(1, 2, -1)$ in the direction of $\mathbf{v} = \langle 3, -1, 2 \rangle$.

$x = 1 + 3t, y = 2 - t, z = -1 + 2t$.

- 9 Find the equation of the plane through $(1, 0, 2)$ with normal vector $\mathbf{n} = \langle 2, -3, 1 \rangle$.
 $2(x - 1) - 3(y - 0) + 1(z - 2) = 0$, i.e. $2x - 3y + z = 4$.

Motion along a curve

- 10 A particle's position is $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$. Find $\mathbf{v}(t)$ and show that the speed $|\mathbf{v}(t)|$ is minimized at $t = 0$.
 $\mathbf{v}(t) = \langle e^t, -e^{-t}, \sqrt{2} \rangle$. $|\mathbf{v}(t)|^2 = e^{2t} + e^{-2t} + 2$. By AM-GM, $e^{2t} + e^{-2t} \geq 2$ with equality at $t = 0$, so $|\mathbf{v}(t)|^2 \geq 4$, minimized (equal to 4) at $t = 0$.

Distance from a point to a line and plane

- 11 Find the distance from the point $P = (1, 2, 3)$ to the line through $(0, 0, 0)$ with direction $\mathbf{v} = \langle 1, 1, 1 \rangle$, using $d = \frac{|\mathbf{w} \times \mathbf{v}|}{|\mathbf{v}|}$ where $\mathbf{w} = \langle 1, 2, 3 \rangle$.
 $\mathbf{w} \times \mathbf{v} = \langle (2)(1) - (3)(1), (3)(1) - (1)(1), (1)(1) - (2)(1) \rangle = \langle -1, 2, -1 \rangle$.
 $|\mathbf{w} \times \mathbf{v}| = \sqrt{1 + 4 + 1} = \sqrt{6}$. $|\mathbf{v}| = \sqrt{3}$. $d = \frac{\sqrt{6}}{\sqrt{3}} = \sqrt{2}$.
- 12 Find the distance from the point $(2, 1, -1)$ to the plane $x + 2y - 2z = 5$, using $d = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$.
 $d = \frac{|1(2) + 2(1) - 2(-1) - 5|}{\sqrt{1 + 4 + 4}} = \frac{|2 + 2 + 2 - 5|}{3} = \frac{1}{3}$.

Curvature and the normal component of acceleration

- 13 For a particle moving with position $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$, find $\mathbf{v}(t) = \mathbf{r}'(t)$ and $\mathbf{a}(t) = \mathbf{r}''(t)$, then find the speed at $t = 1$.
 $\mathbf{v}(t) = \langle 1, 2t, 0 \rangle$, $\mathbf{a}(t) = \langle 0, 2, 0 \rangle$ (constant). Speed at $t = 1$: $|\mathbf{v}(1)| = |\langle 1, 2, 0 \rangle| = \sqrt{5}$.

Projections and angles between vectors

- 14 Find the angle between $\mathbf{a} = \langle 1, 1, 0 \rangle$ and $\mathbf{b} = \langle 1, 0, 1 \rangle$, using $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$.
 $\mathbf{a} \cdot \mathbf{b} = 1$. $|\mathbf{a}| = |\mathbf{b}| = \sqrt{2}$. $\cos \theta = \frac{1}{2}$, so $\theta = \frac{\pi}{3}$ (60°).
- 15 Find the scalar projection of $\mathbf{a} = \langle 3, 4 \rangle$ onto $\mathbf{b} = \langle 1, 0 \rangle$, using the formula $\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$.
 $\mathbf{a} \cdot \mathbf{b} = 3$, $|\mathbf{b}| = 1$: scalar projection $= \frac{3}{1} = 3$.

An applied vector problem

- 16** An airplane's velocity relative to the air is $\mathbf{v}_{air} = \langle 200, 0 \rangle$ km/h (heading due east), and the wind velocity is $\mathbf{v}_{wind} = \langle 10, 30 \rangle$ km/h. Find the airplane's actual ground velocity $\mathbf{v}_{ground} = \mathbf{v}_{air} + \mathbf{v}_{wind}$, and find its ground speed.

$$\mathbf{v}_{ground} = \langle 210, 30 \rangle. \text{ Ground speed} = \sqrt{210^2 + 30^2} = \sqrt{44100 + 900} = \sqrt{45000} \approx 212.1 \text{ km/h.}$$

- 17** A drone hovers with position vector $\mathbf{r}(t) = \langle 5\cos t, 5\sin t, 2 + 0.5t \rangle$ (meters, seconds), tracing a rising spiral. Find the drone's velocity vector $\mathbf{v}(t)$, then find the drone's speed at $t = \frac{\pi}{2}$, and separately find its height above the ground at that same instant.

$$\mathbf{v}(t) = \langle -5\sin t, 5\cos t, 0.5 \rangle. \text{ Speed: } |\mathbf{v}(t)| = \sqrt{25\sin^2 t + 25\cos^2 t + 0.25} = \sqrt{25.25} \approx 5.02 \text{ m/s}$$

(constant, independent of t). Height at $t = \frac{\pi}{2}$: $z = 2 + 0.5(\frac{\pi}{2}) = 2 + \frac{\pi}{4} \approx 2.79 \text{ m.}$

- 18** For $\mathbf{a} = \langle 2, -3, 1 \rangle$ and $\mathbf{b} = \langle -1, 2, 4 \rangle$, find each of the following:

a $2\mathbf{a} - \mathbf{b}$ $2\langle 2, -3, 1 \rangle - \langle -1, 2, 4 \rangle = \langle 4, -6, 2 \rangle - \langle -1, 2, 4 \rangle = \langle 5, -8, -2 \rangle.$

b $\mathbf{a} \times \mathbf{b}$ $\mathbf{a} \times \mathbf{b} = \langle (-3)(4) - (1)(2), (1)(-1) - (2)(4), (2)(2) - (-3)(-1) \rangle = \langle -14, -9, 1 \rangle.$

c The area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ Area
 $= |\mathbf{a} \times \mathbf{b}| = \sqrt{196 + 81 + 1} = \sqrt{278} \approx 16.67.$

- 19** A particle's position is $\mathbf{r}(t) = \langle t^2 - 1, 2t, t^3 \rangle$. Find each of the following at $t = 1$:

a The velocity vector $\mathbf{v}(1)$. $\mathbf{v}(t) = \langle 2t, 2, 3t^2 \rangle$: $\mathbf{v}(1) = \langle 2, 2, 3 \rangle.$

b The acceleration vector $\mathbf{a}(1)$. $\mathbf{a}(t) = \langle 2, 0, 6t \rangle$: $\mathbf{a}(1) = \langle 2, 0, 6 \rangle.$

c The speed $|\mathbf{v}(1)|$. $|\mathbf{v}(1)| = \sqrt{4 + 4 + 9} = \sqrt{17}.$